

# GE23131-Programming Using C-2024

Quiz navigation



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| Status    | Finished                           |
| Started   | Monday, 23 December 2024, 5:33 PM  |
| Completed | Tuesday, 26 November 2024, 8:19 PM |
| Duration  | 26 days 21 hours                   |

Question 1

Correct

Marked out of 3.00

Flag question

Write a program that prints a simple chessboard.

Input format:

The first line contains the number of inputs T.  
The lines after that contain a different values for size of the chessboard

Output format:

Print a chessboard of dimensions size \* size. Print a Print W for white spaces and B for black spaces.

Input:

2  
3  
5

Output:

WBW  
BWB  
WBW  
WBWBW  
BWBWB

WBWBW

BWBWB

WBWBW

**Answer:** (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main()
3  {
4      int T, d, i = 0, i1, i2, o;
5      char c;
6      scanf("%d", &T);
7      while(i < T)
8      {
9          scanf("%d", &d);
10         i1 = 0;
11         while(i1 < d)
12         {
13             o = 1;
14             i2 = 0;
15             if(i1 % 2 == 0)
16             {
17                 o = 0;
18             }
19             while(i2 < d)
20             {
21                 c = 'B';
22                 if(i2 % 2 == o)
23                 {
24                     c = 'W';
25                 }
26                 printf("%c", c);
27                 i2++;
28             }
29             i1+=1;
30             printf("\n");
31         }
32         i = i + 1;
33     }
34     return 0;
35 }
```

|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 2     | WBW      | WBW   | ✓ |
|   | 3     | BWB      | BWB   |   |
|   | 5     | WBW      | WBW   |   |
|   |       | WBWBW    | WBWBW |   |
|   |       | BWBWB    | BWBWB |   |
|   |       | WBWBW    | WBWBW |   |

|  |  |       |       |  |
|--|--|-------|-------|--|
|  |  | BWBWB | BWBWB |  |
|  |  | WBWBW | WBWBW |  |

Passed all tests! ✓

Question **2**

Correct

Marked out of  
5.00

🚩 Flag question

Let's print a chessboard!

Write a program that takes input:

The first line contains T, the number of test cases

Each test case contains an integer N and also the starting character of the chessboard

Output Format

Print the chessboard as per the given examples

Sample Input / Output

Input:

2

2 W

3 B

Output:

WB

BW

BWB

WBW

BWB

**Answer:** (penalty regime: 0 %)

```
1 #include<stdio.h>
```

```

2 int main()
3 {
4     int T, d, i, i1, i2, o, z;
5     char c, s;
6     scanf("%d", &T);
7     for(i = 0; i < T; i++)
8     {
9         scanf("%d %c", &d, &s);
10        for(i1 = 0; i1 < d; i1++)
11        {
12            z = (s == 'W') ? 0 : 1;
13            o = (i1 % 2 == z) ? 0 : 1;
14            for(i2 = 0; i2 < d; i2++)
15            {
16                c = (i2 % 2 == o) ? 'W' : 'B';
17                printf("%c", c);
18            }
19            printf("\n");
20        }
21    }
22    return 0;
23 }

```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 2     | WB       | WB  | ✓ |
|   | 2 W   | BW       | BW  |   |
|   | 3 B   | BWB      | BWB |   |
|   |       | WBW      | WBW |   |
|   |       | BWB      | BWB |   |

Passed all tests! ✓

Question 3

Correct

Marked out of  
7.00

🚩 Flag question

Decode the logic and print the Pattern that corresponds to given input.

If N= 3

then pattern will be :

10203010011012

\*\*4050809

\*\*\*\*607

If N= 4, then pattern will be :

If  $N = 4$ , then pattern will be:

```
1020304017018019020
```

```
**50607014015016
```

```
****809012013
```

```
*****10011
```

### Constraints

$2 \leq N \leq 100$

### Input Format

First line contains  $T$ , the number of test cases

Each test case contains a single integer  $N$

### Output

First line print Case # $i$  where  $i$  is the test case number

In the subsequent line, print the pattern

### Test Case 1

3

3

4

5

### Output

Case #1

```
10203010011012
```

```
**4050809
```

```
****607
```

Case #2

1020304017018019020

\*\*50607014015016

\*\*\*\*809012013

\*\*\*\*\*10011

Case #3

102030405026027028029030

\*\*6070809022023024025

\*\*\*\*10011012019020021

\*\*\*\*\*13014017018

\*\*\*\*\*15016

**Answer:** (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main(){
3      int n, v, p3, c, in, i, i1, i2, t, ti;
4      scanf("%d", &t);
5      for(ti = 0; ti < t; ti++){
6          v = 0;
7          scanf("%d", &n);
8          printf("Case #%d\n", ti + 1);
9          for(i = 0; i < n; i++){
10             c = 0;
11             if(i > 0){
12                 for(i1 = 0; i1 < i; i1++) printf("");
13             }
14             for(i1 = i; i1 < n; i1++){
15                 if(i > 0) c++;
16                 printf("%d0", ++v);
17             }
18             if(i == 0){
19                 p3 = v + (v * (v - 1)) + 1;
20                 in = p3;
21             }
22             in = in - c;
23             p3 = in;
24             for(i2 = i; i2 < n; i2++){
25                 printf("%d", p3++);
26                 if(i2 != n - 1) printf("0");
27             }printf("\n");
28         }
29     }
30 }
```

| Input | Expected | Got |
|-------|----------|-----|
|       |          |     |

|   |   |                          |                          |   |
|---|---|--------------------------|--------------------------|---|
| ✓ | 3 | Case #1                  | Case #1                  | ✓ |
|   | 3 | 10203010011012           | 10203010011012           |   |
|   | 4 | **4050809                | **4050809                |   |
|   | 5 | ****607                  | ****607                  |   |
|   |   | Case #2                  | Case #2                  |   |
|   |   | 1020304017018019020      | 1020304017018019020      |   |
|   |   | **50607014015016         | **50607014015016         |   |
|   |   | ***809012013             | ***809012013             |   |
|   |   | *****10011               | *****10011               |   |
|   |   | Case #3                  | Case #3                  |   |
|   |   | 102030405026027028029030 | 102030405026027028029030 |   |
|   |   | *6070809022023024025     | *6070809022023024025     |   |
|   |   | ****10011012019020021    | ****10011012019020021    |   |
|   |   | *****13014017018         | *****13014017018         |   |
|   |   | *****15016               | *****15016               |   |

Passed all tests! ✓

Finish review